

**Human Factors in Telemedicine: A Complex Systems Advisory for Enhancing Design,
Adoption, and Operational Efficiency**

Asha Golveo

Embry-Riddle Aeronautical University

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Dr. Edwin Odisho

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A complex sociotechnical system (CSTS) consists of interdependent elements of people, technology, organizations, procedures, and environments. Telemedicine platforms are a CSTS because they integrate technological innovation with human, organizational, and cultural dimensions of healthcare delivery. The adoption of telemedicine was propelled by the need for remote options during the COVID-19 pandemic, its rapid implementation resulting in significant changes to clinical workflows, and impacting how healthcare providers operate and communicate with their patients (Bonet Olivencia & Sasangohar, 2023). Telemedicine is a dynamic system comprising multiple interdependent elements that collectively influence safety, efficiency, and quality of care. Effective implementation and performance of telemedicine requires teams collaborating in a complex, distributed network, while navigating regulatory environments, unfamiliar technology, and varying levels of digital literacy (Haleem et al., 2021).

System Components

The system components of telemedicine extend beyond the hardware and software platforms that enable remote communication. Telemedicine encompasses electronic health records (EHR), diagnostic devices, data transmission protocols, and security frameworks, as well as human operators, workflows, and decision-making practices. Each component interacts with others in both linear and nonlinear ways, generating new opportunities for accessibility and coordination, but also new risks and vulnerabilities. The associated disciplines involved in telemedicine reflect its complexity, with physicians and nurses providing the clinical foundation, information technology and computer science developing the digital infrastructure, systems

engineering and human factors shaping usability and safety, and law and policy establishing boundaries of compliance.

People Components

The patient and provider are the central individuals in the telemedicine system, with all surrounding tasks focused on ensuring effective healthcare delivery between them. Their willingness to embrace this technology plays a crucial role in the platform's success. Powell et al. (2017) found that many patients perceived virtual health appointments as convenient and efficient, allowing them to see providers and order medication from home. However, patients have varying levels of digital literacy, and many have concerns about privacy. Although patients living in remote or insecure locations may benefit most from virtual care, technical challenges and poor internet connection can hinder their use of telemedicine.

Healthcare providers can also be a barrier to telemedicine acceptance if they perceive training as insufficient, scheduling and coordination is complex, or systems are disruptive to their workflow (Zhang et al., 2021). Providers in telemedicine are defined as healthcare physicians, nurses, or administrative and support staff. Physicians and nurses have an interdependent relationship requiring collaboration and communication, while sharing the responsibility of effectively providing quality patient care, despite physical distance. Physicians rely on nurses to complete patient intake, providing them essential information to make informed decisions. Critical to the success of telemedicine is fostering trust, respect, and mutual understanding between physicians and nurses (Aseeri et al., 2024). However, communication failures and misunderstandings can impede collaboration and amplify the risks for medical errors.

The telemedicine platform is built by software engineers, system architects, and data managers who must ensure the system is usable, secure, interoperable, and compliant with regulations. Through an iterative process, developers must respond to clinical feedback and make necessary changes to the system. Technical support staff ensure patients and clinicians can access the system, maintain continuity of the system, and troubleshoot any connectivity issues.

Administrative and support staff are essential to the seamless operation of the work system by handling critical logistics, technical maintenance, and patient-facing functions. The administrative team includes schedulers, medical assistants, and billing staff. They coordinate patient appointments, manage calendars for providers and patients, maintain patient records, handle insurance billing, and act as a liaison between patients and providers. Support staff can improve the patient experience by handling administrative duties, thereby allowing providers to focus on delivering quality patient care.

Organizational leaders and policy makers can indirectly alter reimbursement models in healthcare, thereby affecting equitable access to telemedicine. Hospital administrators establish policies and determine how to allocate resources, thereby shaping how telemedicine is integrated into clinical workflows. Furthermore, government regulators and accreditation bodies establish quality and safety standards for healthcare organizations, dictating how and where telemedicine can operate. They indirectly form part of the human component by developing accountability frameworks in which providers and patients must operate. Enacted policies can influence the function and integration of system boundaries, ultimately creating or removing complexities that impact compliance and reimbursement. For example, only 43 states have laws regulating private payer reimbursement policies for telemedicine, while 21 states have equal payment for the same service (CCHP, 2022).

Technical Components

The tools and technology that encompass telemedicine include hardware devices used to access clinical applications, software programs that operate the system, network infrastructure, and cybersecurity infrastructure. Patients and providers communicate using devices such as laptops, smartphones, or tablets, with built-in or external cameras and microphones. The condition and age of the devices can significantly impact quality of communication, as well as contribute to the complexity and usability of the system. In addition, providers may employ wearable devices and diagnostic tools to continuously monitor their patients remotely. The software infrastructure includes the programs and other operating systems used to run the telemedicine platform, including licensed third-party software, patient portals, and electronic health records (EHR) (Zhang et al., 2021). The design of telemedicine's user interface is critical for patient usability, and should be crafted to match the literacy levels of the patient population (Bonet Olivencia & Sasangohar, 2023). Furthermore, accessibility features such as translation tools, closed captioning, and adaptive devices can ensure that telemedicine is inclusive to patients with disabilities.

Telemedicine platforms must have video conferencing systems that are HIPAA compliant and provide real-time audio and video interaction. In addition, asynchronous messaging within the platform enables providers to securely chat or email with their patients, allowing them to respond when it is most convenient. Network infrastructure that supports telemedicine operations includes broadband connectivity, cloud services, data storage, and middleware for interoperability between systems. Cybersecurity in telemedicine must protect sensitive patient data, preventing unauthorized access and data breaches by utilizing security measures such as encryption and multi-factor authentication (Bocetta, 2024).

Environmental Components

The physical environment in telemedicine includes the setting where the patient and provider are located when engaging in the virtual appointment. Environmental factors in telemedicine include internet infrastructure, a significant challenge to telemedicine impacting both patients and providers. Most patients attend their appointment from home, where privacy, lighting, space, and internet stability can affect the comfort and quality of their interaction (Zhang et al., 2021). Furthermore, noise outside the home, interruptions from childcare, and other responsibilities can interfere with the call. The patient's environment may not be suitable for a medical consultation, especially when discussing sensitive issues that require privacy. When providers conduct the appointment from their home office, they may experience the same challenges as patients regarding space and connectivity. Similarly, providers found difficulty in maintaining awareness of their own surroundings, thereby compromising confidentiality and patient privacy when delivering care from a non-traditional setting (Houser et al., 2023). However, clinical settings also have limited space, and providers may encounter noise and distractions from events inside the hospital. Patients and providers in rural areas lack adequate broadband access, with limited bandwidth resulting in poor video quality, delayed audio, or dropped calls (Bonet Olivencia & Sasangohar, 2023).

System Boundaries and Complexities

Telemedicine operates across multiple system boundaries including organizational, technical, regulatory, cultural, and spatial. These boundaries are beneficial in providing quality assurance and distributed expertise, while allowing for redundancies that improve system

resilience and business continuity. However, these system boundaries can also produce complexities and sources of friction that impact patient quality of care.

Organizational Boundaries

Telemedicine involves the integration of multiple organizations such as hospitals, insurance providers, technology vendors, and pharmacies. Each entity has its own policies, IT infrastructure, data-sharing procedures, and governance structures. They operate independently while being highly interdependent, leading to management challenges that could result in mistakes and inefficiencies when decisions are made in isolation (Sheng et al., 1999). The SEIPS model emphasizes that success for the "organization" component must include teamwork, coordination, and communication (Zhang et al., 2021). Contextual and organizational differences must be identified and addressed to properly manage telemedicine services. If challenges are not mitigated or resolved, organizational complexities can impact everything from initial planning and technology adoption to daily workflow, communication, and long-term sustainability (Wang et al., 2020).

Even within a single organization, implementing telemedicine introduces significant complexities by altering established workflows, team structures, and communication patterns. A telemedicine system must have clear policies to ensure that standard of care is comparable to an in-person visit. Dünnebeil et al. (2012) found that providers viewed standardization and technology utilization as significant drivers when adopting telemedicine. Although the skills and experience of a provider are important to patient care, the work system must also exhibit high design quality to facilitate optimal performance from the health practitioner (Carayon et al., 2006). A provider's performance may vary when working within different organizational

systems. Ford et al. (2023) found that scheduling, lack of EHR interoperability, absence of pre-appointment planning, and issues with information exchange as barriers to performance. Furthermore, provider performance may suffer when they feel overloaded with clinical work, digital record-keeping, and patient communication across multiple platforms. Hospital leadership must take responsibility for providing the necessary resources and funding to alleviate organizational pressure, while fostering a culture that encourages teamwork, and embraces new technology (Martinez et al., 2017).

Technical Boundaries

Data sharing is a primary concern when operating across multiple institutional boundaries, which can raise issues of ownership, privacy, and system interoperability. Lack of interoperability is a significant technical challenge, preventing seamless communication and automated data exchange between organizations. Many telemedicine platforms are difficult to integrate with existing hospital information systems such as EHRs, Picture Archiving Communication Systems (PACS), and patient portals (Bonet Olivencia & Sasangohar, 2023). This fragmentation can force providers to shift between multiple platforms to access patient information, leading to inefficiency and increased workload. Furthermore, the complexity of technology requires that providers are given sufficient training to ensure they feel comfortable and confident using the system. The availability of knowledgeable staff with appropriate technical skills is vital for implementation, troubleshooting, and support of telemedicine equipment (Martinez et al., 2017). When patients or providers experience technical errors, it may cause frustration and abandonment of the system if they are not provided technical assistance.

External technology vendors and local IT teams must collaborate during design and development to ensure successful knowledge transfer, and system continuity. While external vendors often design, maintain, and host telemedicine platforms, a hospital's in-house IT staff will eventually support the hospital's daily technical needs. When external vendors collaborate with IT staff to understand the hospital's workflows and security protocols, successful implementation can result in better adoption and satisfaction of the system (Rinne & Dominis, 2024). In addition, compliance officers and IT security teams may strictly enforce HIPAA/GDPR rules at the expense of user-friendly design, frustrating providers and slowing down workflows. Technical teams must collaborate with clinical operations to better understand their needs, building a telemedicine system that can accommodate security and usability.

Cultural Boundaries

Culture influences human performance and cognition in ways that can either enhance or hinder the effectiveness of virtual care (Mansouri-Rad et al., 2013). Subcultures within healthcare professions may interpret digital workflows differently, patients' cultural backgrounds influence trust and communication styles, and societal attitudes toward technology can affect adoption rates (Nittas et al., 2024). Moreover, the norms of interdisciplinary teams, organizational cultures of safety, and even broader cultural narratives about healthcare equity and privacy all influence how telemedicine systems function in practice.

Within the clinical subculture, healthcare providers have established hierarchies and workflows, often rooted in cultural norms, that can impede open communication and foster power imbalances (Ritunga et al., 2024). In some cultures, physicians hold authoritative power over nurses, causing reluctance from foreign-trained nurses to speak up or challenge authority, a

critical issue when it leads to medication errors and diagnostic delay (Ashwaq et al., 2024).

Cultural and professional backgrounds, including language barriers, can contribute to misinterpreted communication styles and misaligned expectations. As telemedicine requires nurses to play a larger role in patient monitoring, intake, triage, and care coordination, physicians must rely heavily on nurses to convey patient information (Bulto, 2024). When Tan et al. (2023) studied collaboration in telemedicine, they found that communication between physicians and nurses could sometimes lead to breakdowns in trust and collaboration, particularly when efficient dialogue and information substance did not match each other's expectations. This has caused some providers to resist telemedicine due to a perceived loss of autonomy, lack of training, and concerns about misdiagnosis without a physical examination (Moulaei et al., 2023).

Regulatory and Legal Boundaries

Telemedicine operates within a complex web of regulatory and legal boundaries that include professional practice, financial considerations, data protection, and broader governmental oversight (van Dyk, 2014). In the United States, the primary regulatory boundaries for telemedicine involve state licensure requirements, requiring providers to be licensed in the state where the patient is located even for virtual visits (Tatro, 2025). The federal government issued waivers to remove the originating site requirement and geographic restrictions, but this waiver is currently extended only through September 30, 2025 (Livanos, 2024; U.S. Department of Health & Human Services, 2025). Laws such as the Health Information Technology for Economic and Clinical Health (HITECH) Act and the Health Insurance Portability and Accountability Act (HIPAA) were enacted to protect patient health information. However, these rules can add complexity when providers operate across state or national lines, presenting a challenge for providers and patients to exchange information when operating within overlapping federal and

state laws (McGraw & Mandl, 2021). Telemedicine platforms introduce new risks to healthcare systems, particularly concerning cybersecurity and the protection of patient data.

Spatial Boundaries

Telemedicine was designed to overcome the constraints of physical space when delivering healthcare, overcoming geographical distance, physical environment, and organizational separation. Spatial limitations have restricted traditional healthcare from reaching remote and rural patients, creating additional challenges for those with mobility issues. However, availability of physical space for confidential appointments can still pose a challenge for patients and providers. When clinical settings lack sufficient space for telemedicine equipment, providers are unable to conduct comprehensive clinical examinations, thus limiting the types of services that can be effectively delivered (Martinez et al., 2017). Telemedicine is suitable for limited cases of medical conditions and situations, and providers must recognize when a patient would be better served by an in-person appointment.

Human Performance Factors In Telemedicine

Human performance factors, involving both patients and providers, have a significant influence on the safety, functionality, and efficiency of telemedicine systems. These factors include technology usability, communication, training, and the system's integration into clinical workflows. If not sufficiently addressed, these human factors can heighten the risk of medical errors, impede access to care, and disrupt clinical operations.

Technology Usability

The goal of telemedicine is to expand access to healthcare by utilizing technology to remove geographical and socioeconomic barriers (Mechanic et al., 2022). However, effective implementation requires teams of people collaborating in a complex, distributed network, while navigating unfamiliar technology and varying levels of digital literacy (Haleem et al., 2021). Telemedicine can perpetuate inequality if the system fails to account for a digital divide, with users abandoning telemedicine if the system is inaccessible and confusing. Moreover, frustrating interfaces can increase a user's cognitive load, leading to errors and an inability to focus on health issues. Designing systems that are user-friendly includes building interfaces that are simple, consistent, and display appropriate feedback. Interfaces should align with the user's mental model, corresponding with the patient's visual and interaction expectations, as well as the provider's expectations for workflow and process. Error handling should account for prevention and recovery by minimizing areas of friction, while allowing the system to fail safely so that users can recover quickly and easily. In addition, feedback is crucial for teaching users how to navigate telemedicine, helping them understand the system by reinforcing successful actions (Dovetail, 2023). Technology can only realize its potential when there is sufficient user engagement.

Trust and Communication

Trust in technology and the healthcare system, including beliefs held by the patient's social network, can also influence an individual's comfort with telemedicine adoption. Communities with historical distrust of the healthcare system may be hesitant to share sensitive health information over a virtual platform, and patients may fear being seen accessing virtual

care, even in the privacy of their own homes. While language barriers between a provider and patient can be an obstacle during in-person appointments, the problem is exacerbated when visual cues like eye contact and body language are lost or misinterpreted during a video call (Vandecasteele et al., 2024). Telemedicine must utilize a holistic approach that considers societal impacts, perceptions, and cultural structures, as well as training programs that include cultural competency to address language and communication barriers.

Clinical Operations

Effective communication between clinical providers is essential for patient safety and trust. Telemedicine often requires a reallocation of responsibilities and perceived professional roles, leading to redundancy and missed tasks if these changes are not communicated (Tan et al., 2023). Operational inefficiencies can introduce higher workload and extreme stress, with the potential for healthcare workers to experience burnout affecting their mental and physical health. Telemedicine can help with optimizing clinical workflows by streamlining the patient onboarding process, automating patient monitoring, and simplifying scheduling. Leadership can mitigate operational gaps by implementing workflows that require team coordination and efficient information transfer, while fostering a culture of mutual trust and respect to improve employee morale. Foremost, engaging key stakeholders during the initial phase of design, from patients, providers, administrators to regulators, can reveal important considerations and solve “gotchas” before they become larger organizational problems.

Human-Centered Design

While pre-consultation forms are standard in telemedicine, their digital implementation varies in efficiency, accessibility, and design. Improving onboarding forms for patients can

include optimizing for mobile devices, pre-populating patient information to allow patients to quickly review and approve, and integrating logic that intelligently presents questions related to previous answers. These enhancements can improve patient completion, and provide clinicians valuable information before the appointment. To improve accessibility, pre-consultation forms should be inclusive to patients with visual and cognitive impairments. Employing web accessibility standards includes proper HTML semantics, utilizing alternative text for images, employing clear color contrast, proper usage of headings, and enabling keyboard navigation (A11y Collective, n.d.). Designing telemedicine with the end users in mind ensures equal access to essential services for all patients, regardless of their abilities, while also increasing the likelihood that providers will adopt and integrate these technologies into their practice.

Conclusion

The complexity of telemedicine as a sociotechnical system underscores the critical interdependence between technology, people, and organizational structures. As the integration of technology into healthcare continues to expand, it is essential to understand the multifaceted components that contribute to its success and challenges. The adoption of telemedicine during the COVID-19 pandemic demonstrates the system's potential to bridge gaps in healthcare delivery. However, the implementation and ongoing success of telemedicine depend heavily on effective collaboration among patients, providers, administrators, and technology developers. Factors such as system usability, digital literacy, regulatory requirements, and cultural dynamics all influence telemedicine's effectiveness in delivering equitable care. Successful telemedicine systems must prioritize human-centered design, streamline clinical operations, and foster clear communication and trust among all stakeholders. This holistic approach ensures that both

patients and providers can effectively navigate the system, enhancing adoption and sustainability over time.

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